

Theorem 5. The dichotomy of the ring invariants

The algebraic registers (chords, discriminant, moment) against the transcendental In-shell of the Hamiltonian

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The cycloring research program · v1.0.0 · Theorem Monograph Edition

Study	CYCLORING
Study	CYCLORING — the Gamma-periods, the defect chain, and the synchronous breathing of a vortex ring
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Version	1.0.0 — the theorem monograph edition
Date	2026-10-08
Code	cycloring/python/cycloring/ (periods · chain · ring · dynamics · ladder · runner · figures)
Data	cycloring/results/protocols/W1..W7_default.json
License	LicenseRef-Proprietary-Wild8Highlander-1.0

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Abstract

The monograph arranges the classical invariants of the regular N-gon on the algebraic-transcendental scale. The pair-distance product is algebraic: the product of $|z_i - z_j|$ equals $R^{\{N(N-1)/2\}} N^{\{N/2\}}$, and the modulus of the discriminant of the binomial $z^N - \sigma$ equals $N^N |\sigma|^{N-1}$ — both registers are algebraic functions of the radius. The impulse $L = 0$ and the moment $I = N \Gamma R^2$ are algebraic; the power sums are cyclotomic.

The Hamiltonian is the only invariant with a transcendental shell: $H(R) = -(\Gamma^2/2 \pi) [(N(N-1)/2) \ln R + (N/2) \ln N]$. The closed form follows from the sine product of Theorem 2 and is checked against the direct pairwise summation with the error $1e-16$. The dichotomy agrees with the periodic register: the boundary algebraic periods serve the algebraic registers, the interior Γ -periods serve the modulation.

Certificates: W4 — the closed form of H against the direct summation $1e-16$, the distance product $1.3e-16$; W4 also holds the rigid rotation ($1.1e-10$) and the invariant drifts $2.0e-12$.

At a glance

Distance product	the product of $ z_i - z_j = R^{\{N(N-1)/2\}} N^{\{N/2\}}$
Discriminant	$ \text{disc}(z^N - \sigma) = N^N \sigma ^{N-1}$
Impulse and moment	$L = 0$; $I = N \Gamma R^2$ — algebraic
Hamiltonian	$H(R) = -(\Gamma^2/2 \pi) [(N(N-1)/2) \ln R + (N/2) \ln N]$
The shell	the only transcendence — $\ln R$
Certificate	W4: $1e-16$ (the H form), $1.3e-16$ (the product)

1. The algebraic registers

The distance between the vertices is $R|1 - \zeta_N^{j-i}|$, so the product over the pairs splits into the radius power and the chord product: $\prod_{i < j} |z_i - z_j| = R^{N(N-1)/2} \prod_m |1 - \zeta_N^m|^{N/2}$. By the sine product of Theorem 2 the second factor equals $N^{N/2}$ — the register is algebraic in R . The discriminant of the binomial $z^N - \sigma$ equals $(-1)^{N(N-1)/2} N^N \sigma^{N-1}$; the root system of Theorem 1 links it to the squared product of the root differences.

$$\prod_{i < j} |z_i - z_j| = R^{\frac{N(N-1)}{2}} N^{\frac{N}{2}}, \quad |\text{disc}(z^N - \sigma)| = N^N |\sigma|^{N-1}$$

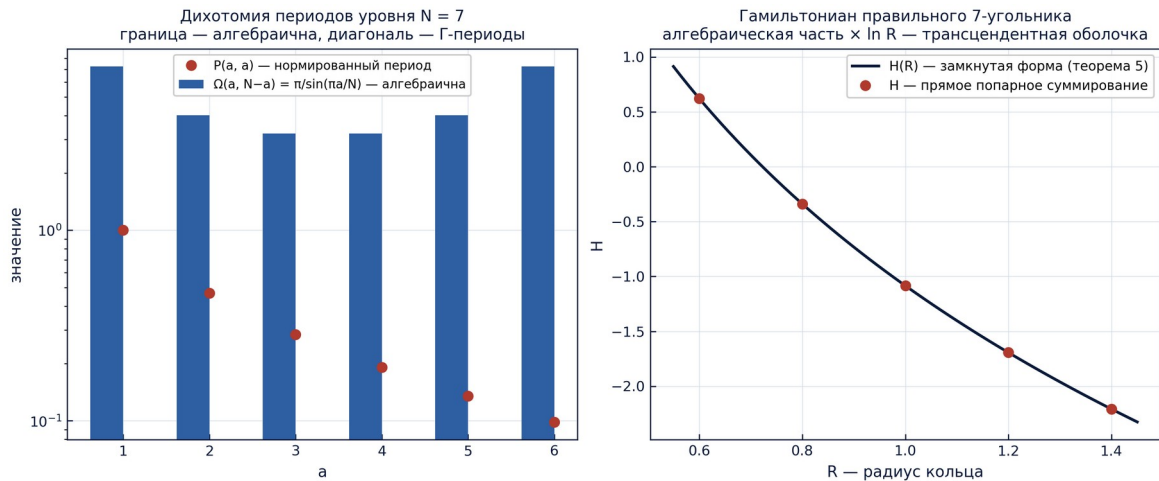
2. The closed form of the Hamiltonian

The logarithm of the product from item 1 is exactly $\sum_{i<j} \ln|z_i - z_j|$. The substitution into $H = - (1/2\pi) \sum_{i<j} \Gamma_i \Gamma_j \ln|z_i - z_j|$ gives the closed form: the term $(N/2) \ln N$ is the constant of the level (the algebraic part inside the shell), and the term with $\ln R$ is the only transcendence. The impulse $L = 0$ is Theorem 1; the moment $I = \sum_k \Gamma^2 |z_k|^2 = N \Gamma^2 R^2$ is a direct computation.

$$H(R) = -\frac{\Gamma^2}{2\pi} \left[\frac{N(N-1)}{2} \ln R + \frac{N}{2} \ln N \right]$$

3. The agreement with the periodic register

The dichotomy of the invariants mirrors the dichotomy of the periods: the boundary algebraic multiples of π (Theorem 2) serve the chords, the discriminant and the moment; the interior Gamma-periods through the witness W_N serve the modulation — the amplitude and the frequency of the breathing (Theorem 3). At the levels $N = 7, 9, 15, 30$ the transducer table ties the two sides together: the higher the level, the stiffer the ring and the closer its invariants stay to the algebraic branch.



The periods and the Hamiltonian: the algebraic part and the transcendental shell.

4. The numerical certificate

Stage W4 ($N = 7$, RK4, two turns): the closed form of H matches the direct pairwise summation with the relative error 10^{-16} ; the distance product — with $1.3 \cdot 10^{-16}$; the measured rotation rate is $1.1 \cdot 10^{-10}$ off the formula; the drifts of H, P, Q, I stay within $2.0 \cdot 10^{-12}$. The protocol is `W4_polygon_flow_default.json`.

The registers of stage W4 (default preset)

register	tolerance	result
rotation rate omega_L	1e-9	1.1e-10
radius drift (shape)	1e-10	6.5e-13
drifts of H, P, Q, I	1e-12	2.0e-12
the closed form of H	1e-12	1e-16
the distance product	1e-12	1.3e-16